**Dictionaries in Python**

A dictionary in Python is a collection of **key-value pairs**. Each key in a dictionary is associated with a value, and you can retrieve or manipulate data using the key. Unlike lists and tuples, dictionaries are **unordered** and **mutable** (changeable).

**1. Creating a Dictionary**

You can create a dictionary using curly braces {} or the dict() function.

**Syntax:**

```
my_dict = {
   "key1": "value1",
   "key2": "value2",
   "key3": "value3"
}
```

**Example:**

Let's create a dictionary of famous cities in Karnataka and their popular dishes.

```
karnataka_food = {
   "Bengaluru": "Bisi Bele Bath",
   "Mysuru": "Mysore Pak",
   "Mangaluru": "Neer Dosa"
}
```

**2. Accessing Dictionary Elements**

To access the values stored in a dictionary, you use the key.

**Example:**

```
print(karnataka_food["Mysuru"])  # Output: Mysore Pak
```

You can also use the get() method to access values, which is safer because it doesn't throw an error if the key doesn't exist.

```
print(karnataka_food.get("Mangaluru"))  # Output: Neer Dosa
print(karnataka_food.get("Shivamogga", "Not Found"))  # Output: Not Found
```

**3. Adding and Updating Dictionary Elements**

You can add new key-value pairs or update existing values in a dictionary.

**Adding an Item:**

```
karnataka_food["Shivamogga"] = "Kadubu"
print(karnataka_food)
```

**Updating an Item:**

karnataka_food["Bengaluru"] = "Ragi Mudde"

**4. Removing Elements from a Dictionary**

You can remove items from a dictionary using several methods:

- pop(): Removes the specified key and returns the associated value.
- mysuru_food = karnataka_food.pop("Mysuru")

print(mysuru_food)  # Output: Mysore Pak

- del: Removes the specified key.

del karnataka_food["Mangaluru"]

- clear(): Empties the dictionary.

karnataka_food.clear()

**5. Dictionary Methods**

Here are some common methods available for dictionaries:

- keys(): Returns all the keys in the dictionary.

print(karnataka_food.keys())  # Output: dict_keys(['Bengaluru', 'Mysuru', 'Mangaluru'])

- values(): Returns all the values in the dictionary.

print(karnataka_food.values())  # Output: dict_values(['Bisi Bele Bath', 'Mysore Pak', 'Neer Dosa'])

- items(): Returns key-value pairs as tuples.

print(karnataka_food.items())  # Output: dict_items([('Bengaluru', 'Bisi Bele Bath'), ('Mysuru', 'Mysore Pak'), ('Mangaluru', 'Neer Dosa')])

- update(): Updates the dictionary with another dictionary or iterable.
- new_dishes = {"Hubballi": "Girmit"}

karnataka_food.update(new_dishes)

**6. Dictionary Characteristics**

- **Unordered**: Dictionary keys are not stored in any particular order.
- **Mutable**: You can change, add, or remove items.
- **Keys Must Be Immutable**: Keys in a dictionary must be of a data type that is immutable, such as a string, number, or tuple.
- **Unique Keys**: A dictionary cannot have duplicate keys. If you try to add a duplicate key, the latest value will overwrite the previous one.

**7. Differences Between Lists, Tuples, Sets, and Dictionaries**

| Feature | List | Tuple | Set | Dictionary |
|---|---|---|---|---|
| **Ordering** | Ordered | Ordered | Unordered | Unordered |
| **Mutability** | Mutable | Immutable | Mutable | Mutable |
| **Duplicates** | Allows duplicates | Allows duplicates | No duplicates | Keys: No duplicates |
| **Indexing** | Supports indexing | Supports indexing | No indexing | Uses keys |
| **Structure** | Indexed collection | Indexed collection | Unordered collection | Key-value pairs |